

# Alexandre Garcia-Duran

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## RESEARCH SUMMARY

PhD student in **computational neuroscience** at the Centre de Recerca Matemàtica (CRM, Barcelona). My research focuses on understanding **vision**: I use **probabilistic graphical models, machine learning, and dynamical systems theory** to study how the brain performs perceptual inference, particularly through the lens of bistable perception. I am also interested in **adversarial robustness** and **connectome-constrained** neural network models, which I explored during a recent visit to the Turaga lab at Janelia Research Campus (HHMI).

## EDUCATION

### PhD Applied Mathematics

2023 – Present

Centre de Recerca Matemàtica (CRM) and Universitat Politècnica de Catalunya (UPC). Computational Neuroscience Group. Supervised by A. Hyafil & M. Molano-Mazón.

### MSc Advanced Mathematics and Mathematical Engineering

2022 – 2023

Universitat Politècnica de Catalunya (UPC).

- Thesis: "An integrative model for perceptual decision-making and movement". Grade 10/10 (Honors).
- GPA 9.1 / 10.

### BEng Biomedical Engineering

2018 – 2022

Universitat de Barcelona (UB).

- Thesis: "Disentangling human decisions under strong time pressure in an expectation-based 2AFC auditory task". Grade 9.7/10.
- Honors in 4 subjects: Internship program, Biomedical Signals & Instrumentation, Biological Systems Modelling, and Biomechanics.
- GPA 8.2 / 10.

## RESEARCH EXPERIENCE

### Doctoral Researcher · Centre de Recerca Matemàtica (CRM)

Nov 2023 – Present

Computational Neuroscience Group. Supervised by A. Hyafil & M. Molano-Mazón.

- Developing **probabilistic graphical models** of visual perception, using **Bayesian inference**.
- Studying how **lateral interactions** in the **visual hierarchy** encode statistical structure for perceptual inference.
- Presented at major conferences: COSYNE, CCN, ICMNS (5 posters + 1 invited talk in 2024–2026).

### Short-term Research Visitor · HHMI Janelia Research Campus

Mar – Apr 2026

Turaga Lab.

- Investigated **adversarial attacks** on optical-flow networks constrained by the *Drosophila* **connectome**.
- Explored computational links between fly visual illusions and network robustness.
- Host: Srinivas C. Turaga.

## Research Assistant · Centre de Recerca Matemàtica (CRM)

Jul 2022 – Oct 2023

Computational Neuroscience Group. Supervised by A. Hyafil, M. Molano-Mazón & G. Huguet.

- Led behavioral analysis and modelling for rat and human sensorimotor data; used **drift-diffusion models**.
- Co-authored a paper published in **Nature Communications** (2024).

## Research Intern · IDIBAPS

Jan – Jun 2022

Brain Circuits & Behavior Lab. Group of Jaime de la Rocha.

- Conducted psychophysics experiments and analyzed human behavioral data (history biases, GLMs).

## Research Intern · Institute for Bioengineering of Catalonia (IBEC)

Jun – Aug 2021

Integrative Cell & Tissue Dynamics lab. Supervised by Nimesh Chahare and Xavier Trepap.

- Worked on mechanobiology and microfluidics experiments.

## PUBLICATIONS

**Garcia-Duran A.**, Wokke M., Pou-Amengual N., Molano-Mazón M., Hyafil A. "Bistable perception reveals a canonical computation in sensory processing". 2026. *In preparation*.

**Garcia-Duran\* A.**, Rodriguez-Garcia\* A., Molano-Mazon M., Hyafil A., Ramaswamy S., Lappalainen J. K. "Measuring Robustness and Efficiency in a Connectome-Constrained Fly Visual System Model on a Collision-Detection Task". *Under review*. 2026. (\* equal contribution)

Molano-Mazón\*, M., **Garcia-Duran\***, A., Pastor-Ciurana\*, J., Hernández-Navarro, Ll., Bektic, L., Lombardo, D., De la Rocha, J., Hyafil, A. "Rapid, systematic updating of movement by accumulated decision evidence". *Nature communications*, 15, 10583; 2024. [DOI](#) (\* equal contribution)

## SELECTED POSTERS & TALKS

### Invited talks

- "Bistable perception reveals a canonical computation in sensory processing." BARCCSYN, Barcelona, May 2026.
- "Bistable perception reveals a canonical computation in sensory processing." Neuroscience Seminars at UPC, May 2026.
- "An attempt to understand (bistable) perception." SIJIMAT seminar, CRM, Barcelona, June 2025.
- "Bistable perception emerges from loopy inference in strongly coupled probabilistic graphs." Neuroscience Seminars at UPC, December 2024.

### Selected Posters (total: 10)

- COSYNE 2026 (Lisbon, Mar 2026). "Lateral connections in visual hierarchy encode statistical structure to support perceptual inference."
- CCN 2025 (Amsterdam, Aug 2025). "Bistable perception emerges from loopy inference in strongly coupled probabilistic graphs."
- COSYNE 2024 (Lisbon, Feb 2024). "Rapid, systematic updating of movement by accumulated decision evidence."
- IBRO World Congress of Neuroscience 2023 (Granada, Sep 2023). "Fast and tight control of motor vigor by accumulated decision evidence."

## FELLOWSHIPS & GRANTS

### PhD Fellowship · Spanish Ministry of Science and Innovation

Nov 2023 – Nov 2027

"Deciphering the brain mechanisms for perceptual inference through bistable perception."

### Computational Resource Grant, National Science Foundation (NSF) ACCESS Program

2026 – 2027

"FlyAdVis: Adversarial robustness and motion detection disruptions in a connectome-constrained fly visual system model."

Project ID: CIS261112

Allocation: 200K ACCESS Credits (~3000 GPU hours)

## TECHNICAL SKILLS

- |                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                            |
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| <ul style="list-style-type: none"><li>• Python (primary — NumPy, PyTorch, scikit-learn)</li><li>• MATLAB · R</li><li>• Bayesian networks / Probabilistic ML</li><li>• Neural Networks (CNN, ANN, LLM)</li><li>• Adversarial attacks &amp; robustness analysis</li></ul> | <ul style="list-style-type: none"><li>• Connectome-constrained neural networks</li><li>• Dynamical systems</li><li>• Psychophysics experimental design</li><li>• Optical flow &amp; image processing</li><li>• Variational autoencoders &amp; Boltzmann Machines</li></ul> |
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## STUDENT SUPERVISION

Supervised two master students (more details on request).

## TRAINING & CERTIFICATES

- CAJAL Advanced Neuroscience Training Programme ([NeuroAI](#)) Neuroscience and AI. Lisbon, Portugal. July 2025.
- Barcelona Summer School for Advanced Modelling of Behaviour ([BAMB!](#)). Barcelona, Spain. July 2024.
- Probabilistic Graphical Models Specialization: Coursera / Stanford Online (2024).
- Deep Learning Specialization: Coursera / DeepLearning.AI (2021).

## COMMUNITY & SERVICE

- Co-organizer of [Neurochats seminars](#) (CRM, Barcelona): an interdisciplinary forum for PhD students and postdocs.
- Co-organizer of [BARCCSYN 2025](#) annual meeting (Barcelona, May 2025).
- [BIYSC](#) teacher. Short lessons on graph theory. July 2024.